

Question ID: 1fe32f7d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$-x^2 + bx - 676 = 0$

In the given equation, b is a positive integer. The equation has no real solution. What is the greatest possible value of b ?

Correct Answer: 51

Rationale

The correct answer is **51**. A quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solution if and only if its discriminant, $-4ac + b^2$, is negative. In the given equation, $a = -1$ and $c = -676$. Substituting -1 for a and -676 for c in this expression yields a discriminant of $b^2 - 4(-1)(-676)$, or $b^2 - 2,704$. Since this value must be negative, $b^2 - 2,704 < 0$, or $b^2 < 2,704$. Taking the positive square root of each side of this inequality yields $b < 52$. Since b is a positive integer, and the greatest integer less than **52** is **51**, the greatest possible value of b is **51**.